ASSIGNMENT : 11

TITLE : Write a Java program to demonstrate the implementation of interfaces.

Problem Statement : Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives : To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory : An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Exercise :

1. Create an interface **Printable** and implement it in Student and Employee classes.

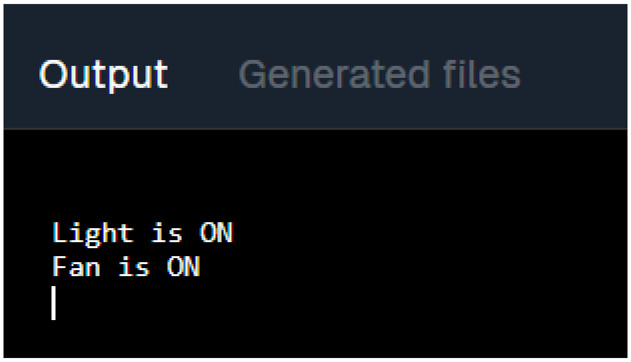
```
interface Printable {
    void print();
}

class Student implements Printable {

    String name = "Anushree";
    int rollNo = 41;

    public void print() {
        System.out.println("Student Details");
        System.out.println("Name: " + name);
        System.out.println("Roll No: " + rollNo);
    }
}

class Employee implements Printable {

    String name = "Anish";
    int empId = 40;

    public void print() {
        System.out.println("Employee Details");
        System.out.println("Name: " + name);
        System.out.println("Employee ID: " + empId);
    }
}

public class Main {
    public static void main(String[] args) {

        Student s = new Student();
        Employee e = new Employee();

        s.print();
        System.out.println();
        e.print();
    }
}
```

Output   Generated files

```
Student Details
Name: Anushree
Roll No: 41

Employee Details
Name: Anish
Employee ID: 40
```

2. Create an interface **Switchable** with a method turnOn(). Implement the interface in **Light** and **Fan** classes to display the device status.

```java
interface Switchable {
    void turnOn();
}

class Light implements Switchable {

    public void turnOn() {
        System.out.println("Light is ON");
    }
}

class Fan implements Switchable {

    public void turnOn() {
        System.out.println("Fan is ON");
    }
}

public class Main {
    public static void main(String[] args) {

        Switchable light = new Light();
        Switchable fan = new Fan();

        light.turnOn();
        fan.turnOn();
    }
}
```



Outcome :

- Understood the concept and implementation of interfaces in Java.
- Achieved abstraction and multiple inheritance of type using interfaces.
- Developed Java programs demonstrating interface implementation in real-world scenarios.

Conclusion :

This assignment demonstrated the implementation of interfaces in Java. Interfaces define a common contract that multiple classes can implement, promoting abstraction, loose coupling, code reusability, and flexibility in object-oriented programming.